

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1213

Glenham 100 kV
December 27, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1213. The Interconnection Customer proposes a 117.1 MW solar photovoltaic facility interconnecting at the Glenham 100 kV substation in the Ohio area. The request was received on May 24, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 373 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$5,060,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mantua - Ellsworth 100 KV	El Paso - Ravenna 500 KV	120.6	93	6.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mantua - Ellsworth 100 KV	Rebuild line with higher-capacity conductor	\$23,812,000	21% (\$5,060,000)
	Total allocated to project		\$5,060,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.